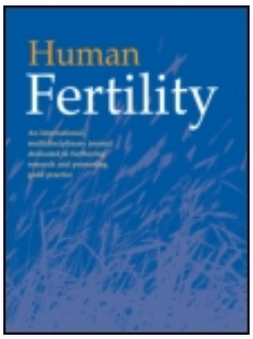

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ORIGINAL ARTICLE

Do women with recurrent miscarriage constitute a high-risk obstetric population?

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ABSTRACT

The aim of this study was to examine if women with history of recurrent miscarriage have a higher risk of maternal and foetal complications in future pregnancies. This was a retrospective case control study that analysed data collected prospectively between 2001 and 2007 from 400 women with history of recurrent miscarriage who achieved pregnancies progressing beyond 24 weeks gestation compared to 39,860 deliveries from the general obstetric database within the same time period. Results showed that women with recurrent miscarriage had significantly increased odds of low Apgar scores at one (odds ratios (OR) 1.57, 95% CI 1.20–2.05) and five minutes (OR 2.0, 95% CI 1.23–3.27), small for gestational age (OR 1.96, 95% CI 1.12–3.43), preterm delivery (OR 1.64, 95% CI 1.22–2.19) and antepartum haemorrhage (OR 7.67, 95% CI 4.23–13.91). The risks were increased in the presence of a male foetus but no difference was observed between primary and secondary miscarriage patients. In conclusion, women with recurrent miscarriage have an increased risk of several maternal and foetal complications and therefore may require closer monitoring during the antenatal period particularly when pregnant with a male foetus.

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Miscarriage; obstetric outcomes; recurrent miscarriage

Introduction

Women with recurrent miscarriage represent an obstetric population that requires dedicated medical and supportive care during the early stages of their pregnancies. The condition is associated with multiple disorders such as congenital and acquired thrombophilias as well as genetic and anatomical abnormalities that require intervention but in many cases the cause of the problem remains unexplained (Jauniaux, Farquharson, Christiansen, & Exalto, 2006). Many of these women will go on to carry a pregnancy beyond the point of foetal viability, but the question remains as to whether they remain at risk during the later stages of pregnancy.

Despite the importance of this group of women, the evidence generated by previous studies remains incomplete. Early studies have produced conflicting evidence, with some studies suggesting poorer obstetric performance (Reginald et al., 1987; Thom, Nelson, & Vaughan, 1992) and others failing to show such an effect (Hughes, Hamilton, & Tulandi, 1991). Furthermore, some studies have used different recurrent miscarriage populations with the inclusion of women with two rather than three consecutive miscarriages (Sheiner, Levy, Katz, & Mazor, 2005).

As highlighted in a literature review (van Oppenraaij et al., 2009), studies on recurrent miscarriage are often subject to bias due to methodological problems such as small population sizes and lack of stratification. The need for further analysis with focus on a homogenous controlled recurrent miscarriage population, including subgroup analyses for different confounding variables, therefore remains.

Over 10 years ago a study from our centre reporting on pregnancy outcomes in a cohort of 162 patients with recurrent miscarriage (Jivraj et al., 2001) demonstrated that these patients have significantly increased prevalence of preterm labour (PTL), small for gestational age (SGA), perinatal loss and caesarean section. Since then, little further evidence has been added to this area. In the current study, we report on our further and more detailed observations on the foetal and maternal outcomes of women with recurrent miscarriage compared to the general obstetric population in a larger cohort of women.

Materials and methods

This was a retrospective analysis of prospectively collected data from the recurrent miscarriage and

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obstetric databases at the Jessop Wing, Royal Hallamshire Hospital between 2001 and 2007.

Recurrent miscarriage patients (study group)

400 consecutive women who had attended the recurrent miscarriage clinic at the Jessop Wing, Royal Hallamshire Hospital and proceeded to achieve pregnancies that progressed beyond 24 weeks gestation between January 2001 and July 2007 were identified from the recurrent miscarriage database and included in the analysis.

Recurrent miscarriage was diagnosed as the occurrence of three consecutive spontaneous miscarriages before 20 weeks gestation (Tuckerman, Laird, Stewart, Wells, & Li, 2004). All subjects underwent investigations according to an established clinical protocol (Li, Makris, Tomsu, Tuckerman, & Laird, 2002), including: prothrombotic studies and antiphospholipid antibody screen, maternal and paternal karyotyping, thyroid function tests and thyroid antibodies, androgen profile (total serum testosterone, sex hormone binding globulin and free androgen index), day two follicle stimulating hormone, luteinizing hormone (LH) and oestradiol, transvaginal ultrasonography, hysterosalpingography and LH-timed Pipelle endometrial biopsy. Hysteroscopy was only performed if an abnormality was suspected on pelvic ultrasound or hysterosalpingography. Treatment of women with recurrent miscarriage is guided by our standard clinical protocol and depends on the cause. Women with unexplained miscarriage are treated conservatively and no interventions are used on an empirical basis. Those with antiphospholipid syndrome are treated with enoxaparin 20 mg once daily and 75 mg of aspirin daily throughout pregnancy. Uterine anomalies that are treated surgically include submucous fibroids, septate uterus and Ashermans syndrome. Women with polycystic ovarian syndrome and associated anovulation are treated with ovulation induction, medically or surgically as the clinical need dictates. This analysis was part of an ongoing service evaluation that was approved by the local audit and clinical effectiveness unit.

Control group

This consisted of a total of 39,860 patients from the general obstetric database (Protos evolution) at the Jessop Wing, Royal Hallamshire Hospital delivering during the same time period. This database is comprehensive and includes all births both at the hospital and at home. Babies admitted to the special care baby unit (SCBU) from other units and twins were excluded from this analysis so as not to bias the results.

Outcome measures

Maternal outcomes

The following outcomes were recorded: (i) Antepartum haemorrhage (APH) (vaginal bleeding from 24 weeks gestation prior to birth of the baby); (ii) Gestational Diabetes (hyperglycaemia first diagnosed in pregnancy according to the WHO criteria; fasting venous plasma glucose level >9 mmol/L or a two hours level of >11 mmol/L after ingestion of 75 g of oral glucose); (iii) Pregnancy induced hypertension (PIH) (diastolic blood pressure of >90 mmHg on two consecutive occasions taken at least four hours apart diagnosed for the first time after 20 weeks gestation); (iv) Preterm labour (PTL) (the number of babies born before 37 completed weeks of gestation) and (v) Emergency caesarean section rate.

Foetal and neonatal outcomes

The following foetal outcomes were recorded: (i) foetal sex ratio (male/female); (ii) the occurrence of intrapartum foetal distress (a pathological heart trace pattern seen during continuous foetal monitoring); (iii) Small for gestational age (SGA) (foetal growth less than two standard deviations below the mean for gestational age according to growth and development record charts); (iv) rate of admission to special care baby unit (SCBU); (v) Stillbirth rate (number of babies born after 24 completed weeks of gestation without signs of life; (vi) Neonatal death rate (death of a baby within 28 days following birth) and (vii) Perinatal death (combination of all stillbirths and neonatal deaths).

All outcomes were classed as primary or secondary as follows:

(1) Primary outcomes: difference in pregnancy outcomes for recurrent miscarriage patients compared to controls.
(2) Secondary outcomes: subgroup analyses were performed to examine:

    (i) differences in pregnancy outcomes for women with unexplained recurrent miscarriage compared to controls;
    (ii) differences in pregnancy outcomes for women with primary recurrent miscarriage compared to primiparous controls delivering for the first time;
    (iii) differences in pregnancy outcomes for women with primary recurrent miscarriage compared to primiparous controls delivering for the first time

(iv) differences in pregnancy outcomes for women with primary recurrent miscarriage compared to primiparous controls delivering for the first time;
(v) differences in pregnancy outcomes for primary versus secondary miscarriage groups; and finally
(vi) effect of the foetal sex on pregnancy outcomes.

Statistical analysis

Data distribution was first explored using histograms and a Kolmogorov–Smirnov test to test for normality. Data were then expressed in the form of mean ± SD, median (range) and percentages as appropriate. For comparisons between two means a *t* test or Mann–Whitney test was used as appropriate. Statistical significance was defined as $p < 0.05$ and the analysis was performed using the Statistical Package for Social Sciences version 14 (SPSS Inc., Chicago, IL). Differences between groups were also expressed using odds ratios (OR) and 95% CI when appropriate and calculated using RevMan software (Review Manager, Version 4.2 for Windows, The Nordic Cochrane Centre, the Cochrane Collaboration, 2003, Copenhagen, Denmark).

Results

The age of the study group was 32.6 ± 5.7 (mean ± SD) years and women had a median BMI of 24.1 (16.1–49.9). A total of 143 patients had primary miscarriage (36%) and 257 (64%) had secondary miscarriage. The age of the control group was significantly lower than the study group 29.0 ± 6.2 ($p < 0.0001$).

The frequency distribution of all potential risk factors for recurrent miscarriage can be seen in Table 1, where the largest category was that of unexplained recurrent miscarriage (38.0%). Secondary miscarriage patients were significantly older than those with primary miscarriage (31.7 ± 5.7 years versus 34.0 ± 5.4 years) but had similar BMI (23.6 [19–44] kg/m^2 for the primary miscarriage group and 29.1 [23–33] kg/m^2 for the secondary miscarriage group, $p = 0.2$).

Table 1. Causes of recurrent miscarriage.

Diagnosis	*N*	Percent
Unexplained	152	38.0
Fibroid (s)	18	4.5
Antiphopholipid syndrome	61	15.2
Structural uterine anomalies	30	7.5
PCOS	40	10.0
Other clotting abnormalities	6	1.5
Chromosomal anomalies	1	0.3
Multiple causes	92	23.0
Total	400	100.01

Obstetric outcomes

Women with recurrent miscarriage had significantly increased odds of low Apgar scores at one (OR 1.57, 95% CI 1.20–2.05) and five minutes (OR 2.0, 95% CI 1.23–3.27), SGA (OR 1.96, 95% CI 1.12–3.43), preterm delivery (OR 1.64, 95% CI 1.22–2.19) admission to SCBU (OR 1.75, 95% CI 1.29–2.36) and perinatal death (OR 3.08, 95% CI 1.44–6.58). Regarding maternal outcomes, there were significantly increased odds for only APH (OR 7.67, 95% CI 4.23–13.91) (Figure 1).

Subgroup analyses

Women with unexplained recurrent miscarriage

There remained significantly increased odds of low Apgar scores at one (OR 1.59, 95% CI 1.04–2.45) and five minutes (OR 2.18, 95% CI 1.02–4.66), SGA (OR 2.82, 95% CI 1.32–6.04) and preterm delivery (OR 1.62, 95% CI 1.01–2.60). Odds for admission to SCBU and perinatal death were no longer significant. Findings regarding intrapartum foetal distress remained unchanged (Figure 2). As for maternal outcomes, there remained significantly increased odds of APH only (OR 6.71, 95% CI 2.45–18.32) (Figure 2).

Primary recurrent miscarriage versus primiparous controls

Women in the study group had significantly increased odds of PIH (OR 3.79, 95% CI 1.88–7.63), preterm delivery (OR 1.78, 95% CI 1.10–2.80) and admission to SCBU (OR 2.09, 95% CI 1.27–3.41). However, they also had significantly lower odds of intrapartum foetal distress (OR 0.25, 95% CI 0.17–0.39).

Primary versus secondary miscarriage groups

There was no evidence for a significant difference in any of the foetal or maternal outcomes between the two groups. When compared to controls, patients with primary miscarriage had an increased risk of preterm delivery (OR 1.74, 95% CI 1.08–2.80), SGA (OR 3.46, 95% CI 1.69–7.09), intrapartum foetal distress (OR 2.68, 95% CI 1.76–4.08) and admission to the SCBU (OR 1.92, 95% CI 1.18–3.12). Similarly, patients with secondary miscarriage also had an increased risk of PTL (OR 1.58, 95% CI 1.09–2.28), intrapartum foetal distress (OR 1.88, 95% CI 1.32–2.67) and admission to the SCBU (OR 1.66, 95% CI 1.13–2.43) and an increased risk of perinatal death (OR 4.14, 95% CI 1.82–9.39).

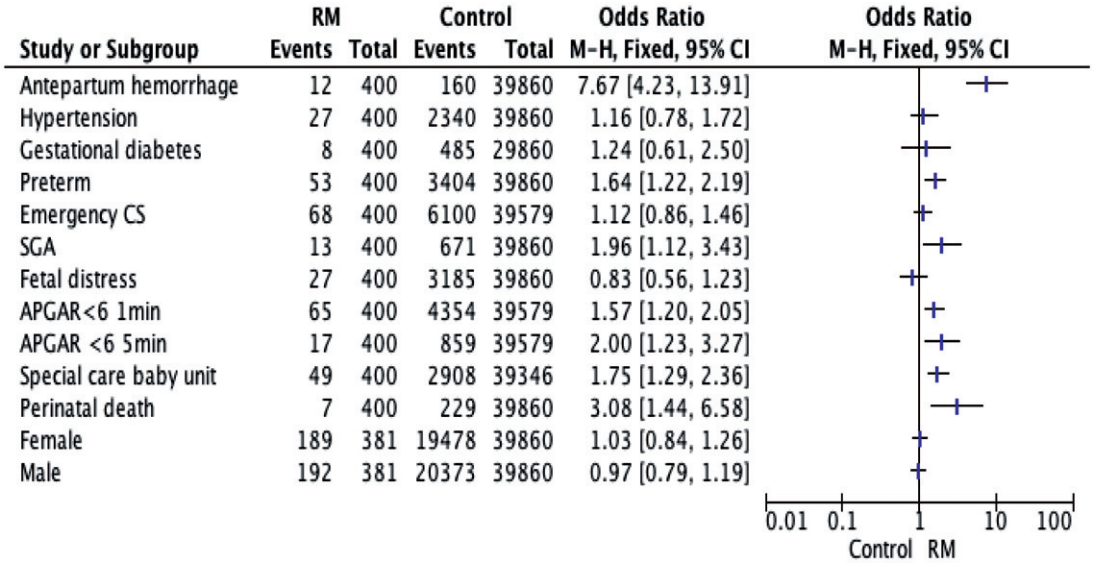

Figure 1. Obstetric outcomes for women with recurrent miscarriage compared to controls.

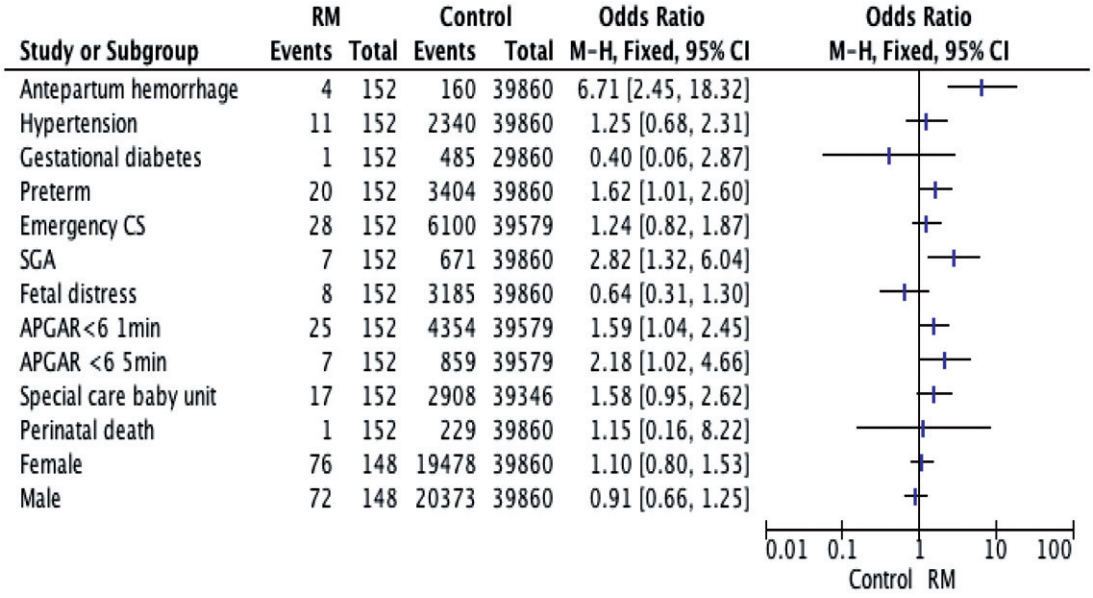

Figure 2. Obstetric outcomes for women with unexplained recurrent miscarriage compared to controls.

Foetal sex

The odds of having a male or female baby was similar in both groups. This was true for the general study population (males OR 0.97, 95% CI 0.79–1.19 and females OR 1.03, 95% CI 0.84–1.26), for patients with unexplained primary miscarriage (males OR 1.28, 95% CI 0.75–2.17 and females OR 0.78, 95% CI 0.46–1.33) and finally for patients unexplained secondary miscarriage (males OR 0.62, 95% CI 0.39–1.0, females OR 1.60, 95% CI 1.01–2.54) (Table 2).

Table 2. Sex ratio in the general population and unexplained miscarriage subgroups.

	Study (*N*)	Control (*N*)	OR [95% CI]
All patients	381	39,860	
Male (*n*)	192	20,373	0.97 [0.79, 1.19]
Female (*n*)	189	19,478	1.03 [0.84, 1.26]
Unexplained primary	56	39,860	
Male (*n*)	32	20,373	1.28 [0.75, 2.17]
Female (*n*)	24	19,478	0.78 [0.46, 1.33]
Unexplained secondary	76	39,860	
Male (*n*)	30	20,373	0.62 [0.39, 1.0]
Female (*n*)	46	19,478	1.60 [1.01, 2.54]

When analysing the whole population, there was evidence for increased odds of preterm delivery (PTL) (OR 2.41, 95% CI 1.28–4.52), intrapartum foetal distress (OR 1.75, 95% CI 1.01–3.05) and admission to SCBU with a male baby (OR 2.08, 95% CI 1.10–3.95). Subgroup analyses showed that male babies still had significantly increased risk of PTL (OR 3.62, 95% CI 1.13–11.65) for the primary recurrent miscarriage group

and for admission to the SCBU for the secondary recurrent miscarriage group (OR 2.58, 95% CI 1.12–5.92). When the results were further limited to women with unexplained recurrent miscarriage, male babies born to mothers of the secondary miscarriage group still had a significantly higher risk of admission to the SCBU (OR 5.88, 95% CI 1.17–29.49). For women with unexplained primary recurrent miscarriage, findings were similar between the groups.

Discussion

This study adds further evidence that women with history of recurrent miscarriage are at higher risk of several foetal and maternal complications during their pregnancy and therefore constitute a higher risk group that requires more intensive antenatal monitoring. The main contribution of the current study is the inclusion of a well-defined recurrent miscarriage group compared to a large obstetric data base as well as the extensive subgroup analyses aimed at minimizing bias.

Our initial analysis of the whole study population demonstrated a significantly increased risk of several foetal and maternal adverse events. However, since recurrent miscarriage can be caused by a heterogeneous group of conditions, some of which are intrinsically associated with an increased risk of pregnancy complications, there is a potential risk of bias due to associated confounding variables. To address this issue, we examined the validity of our results by repeating the analysis in two important subgroups.

The first was a comparison of primary miscarriage patients with the control group limited only to women delivering their first pregnancies. Since the prevalence of some pregnancy complications can vary with parity, comparison of the primary miscarriage group to this selective control group would minimize bias. Results of this analysis showed that the study group still had an increased risk of a number of complications namely PIH, preterm delivery and admission to SCBU. Interestingly, results also showed a decreased risk of foetal distress in the study group. Although the explanation for this finding is not entirely clear, it does suggest that foetal morbidity is mainly related to the higher risk of PIH rather than other causes such as APH.

The second subgroup comparison included only women with unexplained miscarriage. Results remained consistent and showed an increased risk of unspecified APH, PTL, SGA and low Apgar scores. It therefore appears that regardless of the cause of miscarriage, women in this group represent a high-risk obstetric population.

Since all these complications have been linked in previous studies to trophoblastic abnormalities and decidual invasion (Chaddha, Viero, Huppertz, & Kingdom, 2004; Lockwood et al., 2009; Norwitz, 2006; Rutland et al., 2005), we hypothesize that women with recurrent miscarriage may have underlying abnormal placentation. Interestingly though we did not find an increased risk of PIH, another complication related to abnormal placentation, which is similar to previous findings (Jivraj et al., 2001). The reason for this inconsistency is yet to be explained.

The next part of the study examined the effect of the type of miscarriage (primary versus secondary) on pregnancy outcomes. Data from a large epidemiological study examining pregnancy outcomes in a group of women following an initial miscarriage compared to women with one previous successful pregnancy suggested that women with primary miscarriage were at significant risk of developing pre-eclampsia, unspecified APH and low birth weight (Bhattacharya, Townend, Shetty, & Campbell, 2008). Although the current study examines a different population (i.e. women with recurrent miscarriage), it was interesting to examine if the occurrence of a previous healthy pregnancy offered some protective effect.

We found no significant difference between primary and secondary miscarriage groups and both were found to be at increased risk of significant foetal and maternal complications. These findings are slightly different from our original series (Jivraj et al., 2001) where women with secondary miscarriage had a lower risk of PTL compared to their primary miscarriage counterparts. This could be explained by the fact that the current study includes a significantly larger cohort of patients. Women with both primary and secondary miscarriage may therefore benefit from increased surveillance during their antenatal care.

Finally we examined the influence of foetal sex on pregnancy outcomes. There have been several previous reports of poorer pregnancy outcomes with male babies in the general obstetric populations (Aibar, Puertas, Valverde, Carrillo, & Montoya, 2012; Brettell, Yeh, & Impey, 2008; Melamed, Yogev, & Glezerman, 2010; Sheiner et al., 2004; Tan et al., 2004) whereas in women with recurrent miscarriage, a recent study found that women with secondary recurrent miscarriage were more likely to deliver a female baby that were also more likely to be associated with poorer outcomes compared to male babies (Nielsen et al., 2010).

In the current study, however, we found no difference in the sex ratio between the different subgroups and furthermore we found that male babies were associated with poorer outcomes. The fact that this effect

was found at almost all levels of the analysis (groups and subgroups) strengthens these findings. The only group where a significant effect was not found was the unexplained primary miscarriage group. This was the smallest subgroup (24 males and 18 females) and it is therefore possible that the small sample size had led to a loss of statistical power.

Since the Royal Hallamshire Hospital is a large tertiary referral centre, it is possible that the control group is slightly skewed towards higher risk pregnancies. However, even if this were true, then it would minimize rather than accentuate differences between the groups. The fact that the study group was still found to have a higher prevalence of risks, further strengthens our findings.

Potential sources of bias in this study include, first, that as a large tertiary referral centre for recurrent miscarriage, our recurrent miscarriage population may be skewed towards women with identifiable pathologies such as uterine abnormalities rather than women with unexplained miscarriage. Second, we have not been able to present data regarding the distribution of different ethnic groups within our study or control groups. It should be recognized that different areas in the UK have slightly different populations particularly due to ethnic diversity and it is very unlikely that any single hospital, regardless of size, will be representative of the whole of the UK unless the study was performed in a multicentre setting. However, we do not expect that ethnicity had any major effect on our results as relevant pathologies that are more prevalent in certain ethnic groups such as polycystic ovarian syndrome and fibroids, contributed only a small population of our study group. Furthermore the main strength of our study is subgroup analyses, particularly that focussing on the unexplained miscarriage group, which would naturally exclude those with pathologies that may be influenced by ethnicity. Finally, our study group is composed of a relatively high proportion of women with secondary compared to primary recurrent miscarriage. This may be due to the fact that our cohort represents only women with recurrent miscarriage who proceeded to have a pregnancy progressing beyond 24 weeks. It has been previously described that women with secondary miscarriage are more likely to proceed to have a successful pregnancy outcome, and this may explain our data distribution (Quenby & Farquharson, 1993).

Since the unexplained miscarriage group is the best group in which to examine pregnancy outcomes, as the confounding effects of other pathologies is minimized, a multicentre collaborative effort is necessary to achieve a sufficiently large cohort to allow detailed analysis. Moreover, the use of meta-regression analysis may allow for a more detailed analysis of other potential confounders such as age and BMI.

In conclusion, the current study confirms that women with recurrent miscarriage have an increased risk of several maternal and foetal complications. The most significant complications are those of SGA and APH. The occurrence of a previous live birth does not confer any advantage regarding the prognosis in the next pregnancy and both primary and secondary miscarriage patients need to receive similar counselling and care in future pregnancies. Finally, when these women do have a pregnancy that proceeds beyond 24 weeks, those carrying a male foetus may have a higher risk of several complications. It remains to be seen if early identification of foetal sex should dictate how these patients are monitored and managed in their pregnancy and whether or not this would alter their outcomes.

Disclosure statement

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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